

KARTHIK TATINENI

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EDUCATION

B.Tech in Electronics and Communication Engineering

2022 – 2026

Institute of Aeronautical Engineering, Hyderabad | CGPA: 7.2/10

SKILLS

Languages	Python, JavaScript, SQL
Web	FastAPI, Node.js, REST APIs, React, HTML, CSS, Tailwind CSS
AI/ML	YOLO, CNN, TensorFlow, PyTorch, OpenCV
DevOps	Docker, Nginx, Cloudflare, AWS, Grafana
Databases	Firebase, MongoDB, Redis, MySQL, PostgreSQL
Tools	Git, GitHub, Vercel, Postman, WebSocket, Claude Code CLI, Copilot CLI, Ollama LLMs

EXPERIENCE

Independent Software Developer

Jan 2025 – Present

- Designed and deployed **Klvora Fashion Hub** production full-stack e-commerce platform with end-to-end cart, checkout, and order flow; responsive UI serving real customers. [\[live\]](#)
- Built **PV Enterprise IMS** invoice generation, real-time tracking across 100+ SKUs, analytics dashboard, and role-based admin controls. [\[live\]](#)

PROJECTS

Autonomous AI Agentic Team | *Node.js, React, Docker, MCP, WebSocket* | [\[GitHub\]](#)

- Built a multi-agent AI system converting natural-language prompts into production-ready software via coordinated agent workflows and shared memory state.
- Implemented MCP-based inter-agent communication and a real-time WebSocket dashboard for live agent monitoring across parallel tasks.

Distributed College Event Management System | *React, Firebase, Docker, Nginx, Vercel, AWS* | [\[GitHub\]](#) [\[live\]](#)

- Deployed a high-availability event platform handling 1,000+ live registrations with zero downtime across concurrent admin and participant sessions.
- Architected a scalable backend with microservices, Nginx load balancing, and Cloudflare CDN, achieving low response times under peak load.

Real-Time Helmet & Triple-Ride Detection | *Python, YOLOv3, CNN, TensorFlow, PyTorch, OpenCV* | [\[GitHub\]](#) [\[live\]](#)

- Built a computer vision pipeline detecting helmet violations and triple-riding with 90%+ accuracy on test data; optimized to 25+ FPS for real-time traffic surveillance.
- Fine-tuned YOLOv3 on a custom-labeled dataset and implemented automated violation detection for law enforcement alert systems.

PUBLICATIONS

T. Karthik Sai et al. **Helmet Violation and Triple-Ride Detection Using YOLOv3 and CNN**, ICMCER 2026 (Scopus Indexed) | [Certificate](#)

CERTIFICATIONS

- MySQL & Database Management: Create, Manage & Query Databases | [Certificate](#)
- GenAI For Image & Video Creation | [Certificate](#)
- HTML & CSS: The Complete Web Development Guide | [Certificate](#)

EXTRA-CURRICULAR

- Hosted and organised the official **IARE Consortium** event for 1,000+ participants; designed and developed the end-to-end registration platform .
- Core Member and Logistics Head** , NSS led community outreach and logistics for college-level programs.